



User Guide

Delphi App Translation Studio



Version 1.0

Last changed: August 7, 2026

Windows • Delphi VCL and FireMonkey • Win32 and Win64

Table of Contents

Table of Contents	i
1. Welcome	1
1.1 What the Studio changes	1
2. Installation and Startup.....	1
2.1 First launch.....	1
3. The End-to-End Workflow	2
4. Open and Scan a Project	2
4.1 Text that is collected	2
5. Create and Edit a Language Catalog	3
5.1 Translation statuses	3
5.2 Automatic in-place Codex or Claude workflow	4
5.3 Safety, context, and focused review	4
6. Google and DeepL Provider Alternative	5
7. Obtain and Enter a DeepL API Key	5
8. Obtain and Enter a Google API Key.....	6
9. Direct Provider Translation	7
10. Validate and Export.....	7
11. Integrate a Target Application	8
11.1 Runtime files and preferences	8
12. Translate the Studio Itself	8
13. Troubleshooting	9
14. Security and Privacy	10
15. File and Folder Reference	10
16. Provider Reference and Change Notice	10

1. Welcome

Delphi App Translation Studio is an offline-first Windows developer tool for adding localization to existing Delphi VCL and FireMonkey applications. It scans designer-authored forms and Delphi resourcestring declarations, stores the results in an editable development catalog, coordinates automatic in-place translation by Codex or Claude, verifies protected data, validates translation safety, and exports a compact JSON pack for the target application.

The Studio itself is built with FireMonkey, but it works with both VCL and FMX target projects. The supported build targets are Windows Win32 and Win64. macOS, iOS, Android, Linux, C++Builder, and runtime cloud translation are outside the product scope.

Offline by design. Codex or Claude can edit the saved development catalog directly in the developer workspace; no CSV round trip or translation-provider key is required. Google and DeepL remain direct-provider alternatives. A translated target application reads local JSON and never needs Internet access or credentials.

1.1 What the Studio changes

- Scanning is read-only and does not alter the selected project.
- Saving creates project-local Localization\Development catalog files.
- Export creates project-local Localization\Languages runtime packs.
- Integration is a separate explicit operation with preview, verified backup, apply, and restore.
- Language menu items are persisted in the target DFM or FMX so they remain editable in the Delphi IDE.

2. Installation and Startup

The open-source project can be opened and compiled by anyone with Delphi 13 / RAD Studio 13 Florence. No installer is required for development builds. Open DelphiAppTranslationStudio.dproj, choose Win32 or Win64, and build.

Build products are placed under bin\<Platform>\<Configuration>. Compiler units are placed under dcu. The project deliberately does not target mobile or macOS platforms.

2.1 First launch

1. Run DelphiAppTranslationStudio.exe.
2. Confirm that the title reads Delphi App Translation Studio.
3. Use the left workflow panel. The blue selection bar follows Project, Scan, Translate, Validation, Export, Integration, and Provider Settings.

3. The End-to-End Workflow

Step	Purpose	Primary output
1 Project	Open and identify a Delphi project.	Project profile
2 Scan	Extract designated designer text and resourcestrings.	Scan result
3 Translate	Create a catalog and coordinate in-place AI translation.	Protected AI draft catalog
4 Validation	Check completeness and structural safety.	Issue list
5 Export	Create the compact offline pack.	Runtime JSON
6 Integration	Preview/apply runtime and language-menu wiring.	Integrated target project
Provider Settings	Configure Google or DeepL as an alternative.	Direct-provider drafts

4. Open and Scan a Project

1. Choose Project and select Open Delphi Project.
2. Open a .dproj when available; a .dpr is also accepted.
3. Review the detected framework, platforms, form count, and source count.
4. Choose Scan. The Studio reads text DFM/FMX resources and resourcestring declarations.
5. Review the entry count, elapsed milliseconds, breakdown, and result list.

Binary forms. The scanner expects text DFM/FMX resources. If a legacy DFM is binary, use Delphi's normal form conversion facilities before scanning and keep a source-control copy.

4.1 Text that is collected

- Captions, Text values, prompts, hints, headings, and designated list content.
- Memo and string-list content when it represents user-visible text.
- Delphi resourcestring symbols with stable unit-and-symbol keys.

- Locale metadata such as date, time, number, and currency formatting is entered on the Languages page rather than guessed from UI text.

5. Create and Edit a Language Catalog

1. Choose Translate after scanning.
2. Enter the source language, normally en-US.
3. Enter a target locale such as it-IT, de-DE, fr-FR, or es-ES.
4. Enter the language's native name, such as Italiano or Deutsch.
5. Leave direction as ltr unless the target requires right-to-left metadata.
6. Review or edit the date, time, decimal, thousands, and currency fields. Blank fields are initially populated from Delphi TFormatSettings for the target locale.
7. Choose Create / Save.

The catalog is saved under Localization\Development using the project name and target locale. Each target language has its own development catalog. Existing translations are preserved on later scans when the stable key and source text remain unchanged. Changed source text is flagged for review, and removed entries become obsolete rather than disappearing silently.

5.1 Translation statuses

Status	Meaning
Needs translation	No usable target text is present.
AI draft	Codex or Claude produced an in-place draft that passed protected reload.
Machine translated	DeepL or Google produced a draft that requires review.
Imported	CSV supplied target text that requires review.
Edited	A person changed or entered the target text.
Reviewed	A person reviewed the target text.
Approved	The entry is release-approved.
Source changed	The source changed after a translation existed.

Status	Meaning
Excluded	The entry is intentionally omitted.
Obsolete	The source entry no longer exists in the latest scan.
Error	The entry needs correction after a failed operation or check.

5.2 Automatic in-place Codex or Claude workflow

1. Save the development catalog, then choose Begin AI Mode.
2. The Studio creates a pre-AI recovery snapshot, ensures translation-profile.json, writes an instruction file, locks local catalog edits, and copies the exact prompt.
3. Give the prompt to Codex or Claude. The agent reads the catalog and profile and modifies the development JSON directly in place.
4. When the same changed file fingerprint remains stable for two checks, choose Reload AI Work.
5. The Studio compares every protected catalog and entry field with the exact snapshot. Unsafe changes reject the whole reload; allowed translated text, origin, confidence, and review notes are adopted as AI Draft.
6. Resolve only actionable validation exceptions, then export the offline pack. Cancel / Restore returns to the exact pre-AI catalog.

No interchange detour. The agent edits the canonical project-local JSON itself. CSV remains available for translation companies and spreadsheet workflows, but it is not required for Codex or Claude.

5.3 Safety, context, and focused review

- Designer-property entries are applied automatically by the VCL or FMX runtime adapter.
- Pascal resourcestring entries require an explicit TranslateText(Key, Fallback) call at the intended code location. Mark Manual wiring confirmed only after adding and reviewing that call.
- Translation origin is independent of Draft, Reviewed, and Approved status. A valid AI result is never silently approved.
- Only Needs Translation, Source Changed, Error, and existing AI Draft entries are eligible. Machine-translated, Imported, Edited, Reviewed, Approved, Excluded, and Obsolete work is protected.
- A metadata-only AI confirmation may retain an existing translation that remains correct after a source change; changed entries must identify Codex or Claude and record high, medium, or low confidence.
- The terminology profile records application context, audience, tone, formality, protected names, and preferred terminology.

- Validation focuses on placeholders, accelerators, low confidence, explicit AI review notes, inconsistent repeated terms, source changes, and runtime wiring rather than listing every structurally valid AI draft.
- Exact-source translations from other stable keys are suggestions only. The developer must explicitly accept one, and the accepted target remains Edited rather than inheriting approval.

6. Google and DeepL Provider Alternative

When a direct provider is preferred, the Studio supports DeepL API Free, DeepL API Pro, and Google Cloud Translation Basic v2. Provider accounts, billing, quotas, prices, and supported languages are controlled by the provider and may change. Codex/Claude in-place translation does not use these settings.

Control	Behavior
Provider	Select DeepL or Google Cloud Translation.
DeepL API plan	Select API Free or API Pro so the Studio uses the matching endpoint.
API key	Masked field for a new or replacement key.
Remember securely	Stores the key as a Windows Generic Credential.
Session only	Clear Remember before saving; the key is held only until shutdown.
Replace / Save Key	Records the entered key using the selected storage choice.
Test Connection	Sends a small English-to-Italian request.
Remove Key	Deletes stored and session copies for the selected provider.
Timeout / batch	Controls 5-300 second requests and 1-50 strings per request.

7. Obtain and Enter a DeepL API Key

Use a DeepL API account. A normal DeepL Translator subscription is not automatically a DeepL API plan. Confirm that the account includes API Free or API Pro.

1. Open the official DeepL developer site at <https://developers.deepl.com/> and review current API account requirements.

2. Create or sign in to a DeepL account and subscribe to DeepL API Free or DeepL API Pro as appropriate.
3. Open the account's API Keys tab. Create a separate key for this Studio when the account interface permits multiple keys.
4. Copy the key once and keep it out of source code, JSON catalogs, issue trackers, email, and screenshots.
5. In the Studio, choose Provider Settings and select DeepL.
6. Select API Free or API Pro to match the account. Free uses `api-free.deepl.com`; Pro uses `api.deepl.com`.
7. Paste the key into the masked field.
8. Leave Remember securely checked for Windows Credential Manager, or clear it for Use for This Session Only.
9. Choose Replace / Save Key, then Test Connection.
10. If the test fails, verify the plan selection, key status, account quota/billing, firewall, and target-language availability.

DeepL key and authentication reference: <https://developers.deepl.com/docs/getting-started/auth>

DeepL quickstart: <https://developers.deepl.com/docs/getting-started/quickstart>

DeepL multiple-key guidance: <https://developers.deepl.com/docs/multiple-api-keys>

8. Obtain and Enter a Google API Key

Google Basic v2 only. The Studio uses Cloud Translation Basic v2 because it supports API-key authentication. Cloud Translation Advanced v3 uses different authentication and is not implemented.

1. Open Google Cloud Console at <https://console.cloud.google.com/> and sign in.
2. Create a dedicated Google Cloud project or select an existing project intended for this Studio.
3. Attach an active billing account if Google requires billing for the Cloud Translation API.
4. Open APIs & Services, then Library. Find and enable Cloud Translation API.
5. Open APIs & Services, then Credentials. Choose Create credentials and API key. Current Google Cloud policy requires at least one API restriction when creating a key.
6. Under API restrictions, restrict the key to Cloud Translation API. Apply any additional application restriction that is compatible with the developer computer and organization; desktop usage commonly makes website-referrer restrictions unsuitable.
7. Review quotas and budget alerts in Google Cloud before bulk translation.
8. In the Studio, choose Provider Settings and select Google Cloud Translation.
9. Paste the key into the masked field. The DeepL plan control is disabled because it does not apply.
10. Choose persistent Windows Credential Manager storage or session-only use, then Replace / Save Key and Test Connection.

Google Cloud Translation setup: <https://docs.cloud.google.com/translate/docs/setup>

Google API-key creation and management: <https://docs.cloud.google.com/docs/authentication/api-keys>

Google API-key restrictions: <https://docs.cloud.google.com/api-keys/docs/add-restrictions-api-keys>

Cloud Translation authentication: <https://docs.cloud.google.com/translate/docs/authentication>

9. Direct Provider Translation

1. Open or create the target-language catalog.
2. Confirm the source and target language codes.
3. Configure and test a provider under Provider Settings.
4. Choose Google / DeepL Translation on the Translate page.
5. Read the confirmation message showing how many unresolved strings will be sent to the provider.
Cross-key suggestions are never accepted automatically.
6. Choose Yes to begin. Existing complete reviewed or approved work is preserved.
7. Wait for all batches. Transient HTTP 429 and provider server failures receive bounded retries.
8. Review every machine-translated entry for meaning, placeholders, accelerators, product terminology, tone, and available control space.
9. Save the catalog and run Validation.

Cost control. The confirmation count is not a price quote. Provider billing can depend on characters and account terms. Check the provider dashboard and quotas before a large request.

10. Validate and Export

Structural validation checks catalog metadata, required translations, duplicate keys, source changes, Delphi indexed/sequential placeholders, accelerator keys, and excluded/obsolete conditions. Errors block export. Manual resourcestring wiring is reported as a runtime-readiness warning. Linguistic Reviewed and Approved states are reported separately and are not inferred from structural validation.

1. Choose Validation and Run Validation.
2. Open each listed issue and correct the catalog entry or metadata.
3. Repeat until no errors remain.
4. Choose Export and Export Runtime Pack.
5. Record the output path under Localization\Languages.

11. Integrate a Target Application

Integration adds the offline runtime, generated application-specific unit, startup calls, event wiring, and designer-persisted language menu items. It does not add provider access to the target.

1. Close or save the target project in Delphi.
2. Select Integration and confirm the designated language menu component, normally mnuLanguage.
3. Choose Build Integration Plan and review each proposed operation.
4. Choose Generate Preview. Select every changed file and read its line-numbered original/proposed text. Newly generated files are shown in full.
5. After every file has been viewed, check I reviewed every exact change. Apply remains disabled until that confirmation is checked.
6. Choose Apply. The Studio creates and verifies a pre-change backup, writes atomically, and records a restore manifest.
7. Reopen the target project, inspect the DFM/FMX menu items in the IDE, and build Win32 and Win64.
8. Run the target, select a language, restart if required by the application's integration pattern, and verify all forms.
9. Use Restore to recover the recorded pre-integration state if needed.

11.1 Runtime files and preferences

- Deploy JSON packs beside the target executable under Localization\Languages.
- Run Deploy-LanguagePacks.ps1 from the preview package with -ApplicationDirectory pointing to the output folder, or copy the files through the Delphi deployment manager.
- The selected language is stored in %LOCALAPPDATA%\<ApplicationId>\language.ini.
- The executable folder can therefore remain read-only, as it normally is under Program Files.
- VCL startup applies the first form through generated DPR wiring. FMX integration persists an OnCreate handler in the form resource and applies translation after FMX streaming; an existing handler is preserved.

12. Translate the Studio Itself

1. Run the English Studio and select DelphiAppTranslationStudio.dproj.
2. Scan it and create the desired language catalog.
3. Translate, review, validate, and export the pack.
4. Build the self-integration preview for mnuLanguage.
5. Apply the persisted FMX menu-resource update.
6. Close the running Studio, rebuild it, and start the newly built executable.

7. Select the new language. The preference is applied on the next launch.

The running executable is never overwritten. The JSON pack and preference are separate from the executable, and the rebuild supplies the updated menu on the next run.

13. Troubleshooting

Symptom	Likely action
Project framework unknown	Open the .dproj, confirm VCL/FMX units and project metadata, then rescan.
No scan entries	Confirm text DFM/FMX resources and designated user-visible properties.
HTTP 401/403	Replace the key; verify provider plan, API enablement, restrictions, and billing.
HTTP 429	Quota or rate limit reached; wait, reduce batch size, or review provider quota.
DeepL test fails only on one plan	Select API Free or API Pro to match the account endpoint.
Google test fails	Confirm Basic v2 Cloud Translation API is enabled and key API restriction permits it.
Export blocked	Run Validation and correct every error.
Language menu missing	Confirm the designated menu name and rebuild the integration plan.
Preference cannot be found	Check %LOCALAPPDATA%\<ApplicationId>\language.ini, not the executable folder.
Target remains English	Confirm the JSON pack applicationId and locale, deployment folder, selected preference, and startup ApplyTranslation calls.

14. Security and Privacy

- Only confirmed source strings are sent to the selected provider.
- Do not send secrets, personal data, customer data, or regulated information as UI text.
- Remembered keys are Windows Generic Credentials; session keys are not persisted.
- Removing a key affects only the selected provider.
- Provider settings JSON contains no API key.
- Catalogs, runtime packs, deployment packages, logs, backups, and Git should contain no key.
- Rotate a key immediately in the provider console if it may have been exposed, then replace it in the Studio.

15. File and Folder Reference

Location	Contents
Localization\Development	Editable full development catalogs.
Localization\Languages	Compact offline runtime JSON packs.
export	Generated previews and temporary product output.
docs\guides / docs\pdf	Editable guides and companion PDFs.
%LOCALAPPDATA%\DelphiAppTranslationStudio	Studio settings and language preference, but no remembered key.
Windows Credential Manager	Remembered DeepL/Google Generic Credentials.
%LOCALAPPDATA%\<ApplicationId>	Integrated application's selected-language preference.

16. Provider Reference and Change Notice

Provider setup screens and commercial terms can change after this guide is published. Before creating a production key, compare these steps with the official pages listed in Chapters 7 and 8. Prefer a dedicated, restricted key and review the provider dashboard after the first bulk run.

This guide documents the implemented Studio as of August 7, 2026.